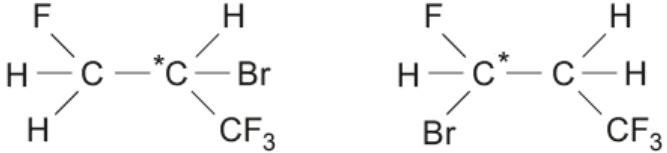
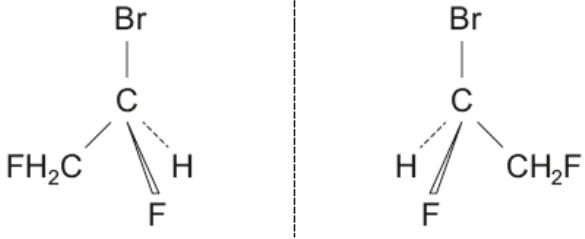


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)			$\text{F}_3\text{CCl} \rightarrow \text{F}_3\text{C}\cdot + \text{Cl}\cdot$		1		1		
	(b)	(i)		the higher the ratio of $\text{HF}:\text{CH}_2\text{Cl}_2$ the higher the yield of CH_2F_2			1	1		
		(ii)		$n(\text{CH}_2\text{F}_2) = 0.0356$ (1) mass of $\text{CH}_2\text{F}_2 = 0.0356 \times 52.02 = 1.85$ (1) ecf possible from incorrect M_r		2		2	1	
	(c)	(i)		high pressure because there are more (gaseous) moles on the left than on the right	1			1		
		(ii)		high temperature as the endothermic reaction needs heat for the position of equilibrium to move to the right	1			1		
		(iii)		separation can be achieved by (fractional) distillation (1) reduce the temperature of the mixture to below -30°C and let the mixture warm up slowly (1,1,1,2-tetrafluoroethane will distil off first and can be condensed in a cold trap) (1)			2	2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		each end of the C=C double bond has two different groups bonded to it	1			1		
		(ii)	I	electrophilic addition	1			1		
			II	(intermediate) carbocations / carbonium ions have similar stabilities / similar reactivities / similar activation energies / are formed at similar rates			1	1		
		(iii)		 correct formulae (1) chiral centres correctly shown on both compounds (1) award (1) for one correct formula and chiral centre		2		2		
		(iv)				1		1		
				Question 8 total	4	6	4	14	1	2